Group 2: Pipeline Workflow

Project Title: Smart Mock Interview Assistant: An NLP-Based System for Evaluating Technical and Behavioural Interview Responses

Team GitHub Repository: https://github.com/ekanth3128/SMART-MOCK-INTERVIEW

Research question: Can neural NLP models accurately evaluate technical and behavioural interview responses and provide personalized feedback in a mock interview simulator for technology-related roles?

1. Full Pipeline Diagram



Figure 1. Full Pipeline Diagram

2. Pipeline Stage Explanation

Stage	Workflow description
Data collection	Collect question/reference-answer data from the three agreed sources. Store original files in /data/raw and record source, date, role, type and licence/usage notes.
Cleaning	Remove duplicate questions, blank records, copied adverts/navigation text, inconsistent spacing and unusable entries.
Preprocessing	Standardise columns such as question_type, question, reference_answer and domain. Preserve grammar for transformer models; apply lowercasing, lemmatisation and stop-word handling only for baseline models where useful.
Model Development	Develop and compare the agreed model approaches. Week 5 work adds a pretrained Sentence-BERT answer similarity component that evaluates candidate answers against reference answers.

Deployment	Package the selected model, preprocessing steps and scoring rubric into a Streamlit application for interactive mock interviews.

Table 1. Pipeline Stage Explanation

3. Feature Engineering and Preprocessing Plan

Area	Plan
Technical text	Clean code-like terms carefully so important phrases such as train-test split, overfitting, precision and recall are not removed.
Behavioural text	Detect STAR structure: Situation, Task, Action and Result. Identify examples, outcomes, reflection and specificity.
Baseline features	TF-IDF, unigrams/bigrams, selected trigrams, answer length and keyword indicators.
Neural features	Token sequences, padded inputs, embeddings and labelled score bands where available or created through rubric-based annotation.
Transformer features	Sentence embeddings and contextual representations from Sentence-BERT, using sentence-transformers/all-MiniLM-L6-v2 for the Week 5 pretrained answer scoring prototype.
Hybrid features	Semantic similarity to reference answer, required concept coverage, clarity/readability and completeness score.

Table 2. Feature Engineering and Preprocessing Plan

4. Model Training, Improvement and Comparison Plan

Model	Role	Purpose and tuning approach
TF-IDF + Logistic Regression/SVM	Baseline	Fast, explainable comparison point; tune n-grams, max features, C value and class weights.
LSTM/BiLSTM/CNN	Neural model	Learns sequence patterns in answers; tune units/filters, dropout, learning rate, batch size and epochs.
Sentence-BERT	Pre-trained similarity model	Uses sentence-transformers/all-MiniLM-L6-v2 to embed the reference answer and user answer, then calculate cosine similarity as a relevance score.
Sentence-BERT	Similarity model	Compares candidate answer with reference answer; useful where direct labelled candidate-answer scores are limited.
Hybrid scoring model	Final candidate system	Combines semantic similarity, keyword coverage, clarity and completeness for realistic interview scoring.

Table 3. Model Training, Improvement and Comparison Plan

Model selection criteria: The chosen approach will be selected using predictive performance, feedback quality, interpretability, fairness across question types, and response speed inside the application.

5. Scoring Framework

Criterion	Description	Weight
Technical correctness	Accuracy of concepts, facts, terminology and reasoning.	30%
Relevance	How directly the response answers the question.	20%
Completeness	Coverage of key expected points or reference concepts.	20%
Clarity	Readable, structured and understandable communication.	15%
Interview readiness	Professional, practical and confident answer quality.	15%

Table 4. Scoring Framework

Behavioural answers will use a related rubric: STAR structure, relevance of example, specificity, reflection, clarity and confidence. The app will return an overall score plus strengths, weaknesses and targeted improvement advice.

6. Model Evaluation Plan

Evaluation area	Metrics and checks
Classification/score bands	Accuracy, precision, recall and macro F1-score. Macro F1 is important if some categories have fewer examples.
Numerical scoring	Mean Absolute Error, Root Mean Squared Error and correlation with human/rubric scores.
Semantic similarity	Compare candidate-reference similarity scores against manually checked answer quality scores. Week 5 checks focus on score range, blank-answer handling, and whether relevant answers score above unrelated answers.
Feedback quality	Human review of whether feedback is relevant, useful, clear, fair and actionable.
Application quality	Measure response time, error rate, usability, stability and whether results are understandable to users.

Table 5. Model Evaluation Plan

7. App Deployment Plan

Preferred route: Frontend app will be used for a fast data science prototype.

```
User opens app → selects role → selects technical/behavioural question → question displayed
→ user types answer → preprocessing pipeline runs → model/scoring framework evaluates answer
→ app displays overall score, category scores, strengths, weaknesses and improvement advice.
```

Listing 1. End-to-End Application Flow

The deployed app will load the saved model, vectoriser/tokeniser, reference answer bank and scoring rules.

The interface will include role selection, question type selection, answer input, scoring output and personalised feedback.

Testing will check model loading, preprocessing, scoring outputs, invalid/empty answers and response time.

8. Project Plan Summary

This workflow provides a complete data science pipeline from dataset collection to model deployment. The system uses interview question datasets to build a mock interview simulator, and Week 5 adds a pretrained Sentence-BERT answer scoring path that compares user answers with reference answers before later feedback and app integration work.